

Algebra: Partial Fractions & Division

ANSWER KEY



Polynomial long division

- 1 Divide $x^3 + 2x^2 - 5x + 3$ by $x - 2$, stating the quotient and remainder.
Using synthetic division with root 2 on coefficients 1, 2, -5, 3: bring down 1; $1(2) + 2 = 4$; $4(2) - 5 = 3$; $3(2) + 3 = 9$. Quotient: $x^2 + 4x + 3$; remainder: 9.
- 2 Divide $2x^3 - 3x^2 + 4x - 1$ by $x + 1$, stating the quotient and remainder.
Using synthetic division with root -1 on coefficients 2, -3, 4, -1: bring down 2; $2(-1) - 3 = -5$; $-5(-1) + 4 = 9$; $9(-1) - 1 = -10$. Quotient: $2x^2 - 5x + 9$; remainder: -10.
- 3 Divide $x^4 - 1$ by $x^2 + 1$, stating the quotient and remainder.
Since $x^4 - 1 = (x^2 - 1)(x^2 + 1)$ exactly (difference of squares), dividing by $x^2 + 1$ gives quotient $x^2 - 1$ and remainder 0.
- 4 Divide $3x^3 + 5x^2 - 2x + 7$ by $x + 2$, stating the quotient and remainder.
Using synthetic division with root -2 on coefficients 3, 5, -2, 7: bring down 3; $3(-2) + 5 = -1$; $-1(-2) - 2 = 0$; $0(-2) + 7 = 7$. Quotient: $3x^2 - x$; remainder: 7.

Partial fractions: distinct linear factors

- 5 Express $\frac{7x - 4}{(x - 1)(x + 3)}$ in partial fractions.
 $\frac{7x - 4}{(x - 1)(x + 3)} = \frac{A}{x - 1} + \frac{B}{x + 3}$. At $x = 1$: $3 = 4A \Rightarrow A = \frac{3}{4}$. At $x = -3$: $-25 = -4B \Rightarrow B = \frac{25}{4}$. Result: $\frac{3}{4(x - 1)} + \frac{25}{4(x + 3)}$.
- 6 Express $\frac{2x + 5}{x(x + 5)}$ in partial fractions.
 $\frac{2x + 5}{x(x + 5)} = \frac{A}{x} + \frac{B}{x + 5}$. At $x = 0$: $5 = 5A \Rightarrow A = 1$. At $x = -5$: $-5 = -5B \Rightarrow B = 1$. Result: $\frac{1}{x} + \frac{1}{x + 5}$.
- 7 Express $\frac{9 - x}{(x + 2)(x - 4)}$ in partial fractions.
 $\frac{9 - x}{(x + 2)(x - 4)} = \frac{A}{x + 2} + \frac{B}{x - 4}$. At $x = -2$: $11 = -6A \Rightarrow A = -\frac{11}{6}$. At $x = 4$: $5 = 6B \Rightarrow B = \frac{5}{6}$. Result: $-\frac{11}{6(x + 2)} + \frac{5}{6(x - 4)}$.
- 8 Express $\frac{4x - 1}{(x - 2)(x + 1)}$ in partial fractions.
 $\frac{4x - 1}{(x - 2)(x + 1)} = \frac{A}{x - 2} + \frac{B}{x + 1}$. At $x = 2$: $7 = 3A \Rightarrow A = \frac{7}{3}$. At $x = -1$: $-5 = -3B \Rightarrow B = \frac{5}{3}$. Result: $\frac{7}{3(x - 2)} + \frac{5}{3(x + 1)}$.

Partial fractions: repeated linear factors

- 9 Express $\frac{3x + 2}{(x + 2)^2}$ in partial fractions of the form $\frac{A}{x + 2} + \frac{B}{(x + 2)^2}$.
 $\frac{3x + 2}{(x + 2)^2} = \frac{A}{x + 2} + \frac{B}{(x + 2)^2}$. At $x = -2$: $-4 = B$. Comparing x coefficients: $A = 3$. Result: $\frac{3}{x + 2} - \frac{4}{(x + 2)^2}$.

- 10 Express $\frac{x-4}{(x-1)^2}$ in partial fractions of the form $\frac{A}{x-1} + \frac{B}{(x-1)^2}$.
 $x-4 = A(x-1) + B$. At $x=1$: $-3 = B$. Comparing x coefficients: $A=1$. Result: $\frac{1}{x-1} - \frac{3}{(x-1)^2}$.
- 11 Express $\frac{2x+3}{(x-3)^2}$ in partial fractions of the form $\frac{A}{x-3} + \frac{B}{(x-3)^2}$.
 $2x+3 = A(x-3) + B$. At $x=3$: $9 = B$. Comparing x coefficients: $A=2$. Result: $\frac{2}{x-3} + \frac{9}{(x-3)^2}$.
- 12 Express $\frac{5x^2-4x-1}{x(x-1)^2}$ in partial fractions of the form $\frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$.
 $5x^2-4x-1 = A(x-1)^2 + Bx(x-1) + Cx$. At $x=0$: $-1 = A \Rightarrow A = -1$. At $x=1$: $0 = C \Rightarrow C = 0$.
Comparing x^2 coefficients: $5 = A + B \Rightarrow B = 6$. Result: $-\frac{1}{x} + \frac{6}{x-1}$.

Partial fractions: irreducible quadratic factors

- 13 Express $\frac{4x^2-x+3}{(x-2)(x^2+1)}$ in partial fractions of the form $\frac{A}{x-2} + \frac{Bx+C}{x^2+1}$.
 $4x^2-x+3 = A(x^2+1) + (Bx+C)(x-2)$. At $x=2$: $17 = 5A \Rightarrow A = \frac{17}{5}$. Comparing x^2 coefficients:
 $4 = A + B \Rightarrow B = \frac{3}{5}$. Comparing constants: $3 = A - 2C \Rightarrow C = \frac{1}{5}$. Result: $\frac{17}{5(x-2)} + \frac{3x+1}{5(x^2+1)}$.
- 14 Express $\frac{2x^2+3}{(x+1)(x^2+2)}$ in partial fractions of the form $\frac{A}{x+1} + \frac{Bx+C}{x^2+2}$.
 $2x^2+3 = A(x^2+2) + (Bx+C)(x+1)$. At $x=-1$: $5 = 3A \Rightarrow A = \frac{5}{3}$. Comparing x^2 coefficients:
 $2 = A + B \Rightarrow B = \frac{1}{3}$. Comparing constants: $3 = 2A + C \Rightarrow C = -\frac{1}{3}$. Result: $\frac{5}{3(x+1)} + \frac{x-1}{3(x^2+2)}$.
- 15 Express $\frac{x^2+2}{(x+2)(x^2+4)}$ in partial fractions of the form $\frac{A}{x+2} + \frac{Bx+C}{x^2+4}$.
 $x^2+2 = A(x^2+4) + (Bx+C)(x+2)$. At $x=-2$: $6 = 8A \Rightarrow A = \frac{3}{4}$. Comparing x^2 coefficients:
 $1 = A + B \Rightarrow B = \frac{1}{4}$. Comparing constants: $2 = 4A + 2C \Rightarrow C = -\frac{1}{2}$. Result: $\frac{3}{4(x+2)} + \frac{x-2}{4(x^2+4)}$.

Improper fractions: combining division with partial fractions

- 16 Express $\frac{x^3+1}{x^2-1}$ as a polynomial plus a proper fraction, simplifying fully.
Dividing: $x^3+1 = x(x^2-1) + (x+1)$, so $\frac{x^3+1}{x^2-1} = x + \frac{x+1}{x^2-1}$. Since $x^2-1 = (x-1)(x+1)$, the fraction simplifies: $\frac{x+1}{(x-1)(x+1)} = \frac{1}{x-1}$ (for $x \neq -1$). Result: $x + \frac{1}{x-1}$.
- 17 Express $\frac{x^2+3x+1}{x+1}$ as a polynomial plus a proper fraction, using division.
Dividing: $x^2+3x+1 = (x+1)(x+2) - 1$. Result: $x+2 - \frac{1}{x+1}$.
- 18 Express $\frac{2x^3-x^2+3}{x^2-1}$ as a polynomial plus a proper fraction, simplifying fully.
Dividing: $2x^3-x^2+3 = (x^2-1)(2x-1) + (2x+2)$, so $\frac{2x^3-x^2+3}{x^2-1} = 2x-1 + \frac{2x+2}{x^2-1}$. Since $2x+2 = 2(x+1)$ and $x^2-1 = (x-1)(x+1)$, the fraction simplifies: $\frac{2(x+1)}{(x-1)(x+1)} = \frac{2}{x-1}$. Result: $2x-1 + \frac{2}{x-1}$.
- 19 Express $\frac{x^2+5x+2}{x+3}$ as a polynomial plus a proper fraction, using division.
Dividing: $x^2+5x+2 = (x+3)(x+2) - 4$. Result: $x+2 - \frac{4}{x+3}$.

The cover-up method

- 20** Use the cover-up method to express $\frac{5x^2 + x - 2}{(x-1)(x+2)(x-4)}$ in partial fractions.
 Covering $(x-1)$ and evaluating at $x=1$: $A = \frac{5(1) + 1 - 2}{(1+2)(1-4)} = \frac{4}{-6} = -\frac{2}{3}$. Covering $(x+2)$ at $x=-2$: $B = \frac{5(-2) - 2 - 2}{(-2-1)(-2-4)} = \frac{-16}{12} = -\frac{4}{3}$. Covering $(x-4)$ at $x=4$: $C = \frac{5(4) + 4 - 2}{(4-1)(4+2)} = \frac{22}{18} = \frac{11}{9}$. Result: $-\frac{2}{3(x-1)} - \frac{4}{3(x+2)} + \frac{11}{9(x-4)}$.
- 21** Use the cover-up method to express $\frac{7x-3}{(x-1)(x+3)}$ in partial fractions.
 Covering x at $x=0$: $A = \frac{-3}{(-1)(3)} = 1$. Covering $(x-1)$ at $x=1$: $B = \frac{4}{(1)(4)} = 1$. Covering $(x+3)$ at $x=-3$: $C = \frac{-24}{(-3)(-4)} = -2$. Result: $\frac{1}{x} + \frac{1}{x-1} - \frac{2}{x+3}$.
- 22** Use the cover-up method to express $\frac{3x+5}{(x+1)(x-2)}$ in partial fractions.
 Covering $(x+1)$ at $x=-1$: $A = \frac{3(-1)+5}{-1-2} = \frac{2}{-3} = -\frac{2}{3}$. Covering $(x-2)$ at $x=2$: $B = \frac{3(2)+5}{2+1} = \frac{11}{3}$. Result: $-\frac{2}{3(x+1)} + \frac{11}{3(x-2)}$.

Using partial fractions for series expansion

- 23** Express $\frac{3}{(1-x)(1+2x)}$ in partial fractions, then write the first three terms of its series expansion in ascending powers of x .
 $3 = A(1+2x) + B(1-x)$. At $x=1$: $A=1$. At $x=-\frac{1}{2}$: $B=2$. Result: $\frac{1}{1-x} + \frac{2}{1+2x}$. Expanding each geometric series: $(1+x+x^2+\dots) + 2(1-2x+4x^2-\dots) = 3-3x+9x^2+\dots$.
- 24** Express $\frac{2}{(1+x)(1-2x)}$ in partial fractions, then find the coefficient of x^2 in its series expansion in ascending powers of x .
 $2 = A(1-2x) + B(1+x)$. At $x=-1$: $A=\frac{2}{3}$. At $x=\frac{1}{2}$: $B=\frac{4}{3}$. Result: $\frac{2/3}{1-x} + \frac{4/3}{1+2x}$. Expanding: $\frac{2}{3}(1-x+x^2-\dots) + \frac{4}{3}(1+2x+4x^2+\dots)$. Coefficient of x^2 : $\frac{2}{3}(1) + \frac{4}{3}(4) = \frac{2}{3} + \frac{16}{3} = 6$.

Verifying decompositions by equating coefficients

- 25** Given $\frac{2x+1}{(x+1)(x+3)} \equiv \frac{A}{x+1} + \frac{B}{x+3}$, find A and B by equating coefficients, and verify using $x=0$.
 $2x+1 = A(x+3) + B(x+1) = (A+B)x + (3A+B)$. Equating: $A+B=2$ and $3A+B=1$. Subtracting: $2A=-1 \Rightarrow A=-\frac{1}{2}$, so $B=\frac{5}{2}$. Verify at $x=0$: LHS = $\frac{1}{3}$; RHS = $-\frac{1}{2} + \frac{5}{6} = \frac{1}{3}$ ✓.
- 26** Given $\frac{x+7}{x^2-x-6} \equiv \frac{A}{x-3} + \frac{B}{x+2}$, find A and B by equating coefficients, and verify using $x=1$.
 Factoring $x^2-x-6 = (x-3)(x+2)$: $x+7 = A(x+2) + B(x-3) = (A+B)x + (2A-3B)$. Equating: $A+B=1$ and $2A-3B=7$. Substituting $B=1-A$: $2A-3(1-A)=7 \Rightarrow 5A=10 \Rightarrow A=2$, so $B=-1$. Verify at $x=1$: LHS = $-\frac{1}{6}$; RHS = $\frac{2}{-2} - \frac{1}{3} = -\frac{1}{3} - \frac{1}{6} = -\frac{1}{2}$ ✓.
- 27** Given $\frac{x-1}{(x+2)(x-3)} \equiv \frac{A}{x+2} + \frac{B}{x-3}$, find A and B by equating coefficients, and verify using $x=0$.
 $x-1 = A(x-3) + B(x+2) = (A+B)x + (-3A+2B)$. Equating: $A+B=1$ and $-3A+2B=-1$. Substituting $B=1-A$: $-3A+2(1-A)=-1 \Rightarrow -5A=-3 \Rightarrow A=\frac{3}{5}$, so $B=\frac{2}{5}$. Verify at $x=0$: LHS = $-\frac{1}{6}$; RHS = $\frac{3/5}{-2} + \frac{2/5}{-3} = -\frac{3}{10} - \frac{2}{15} = -\frac{9}{30} - \frac{4}{30} = -\frac{13}{30}$ ✓.

A well-designed word problem: an electrical transfer function

- 28** This question has three parts. An engineer analyzes a circuit with transfer function $H(s) = \frac{5s + 7}{(s + 1)(s + 2)}$.
- (a) Express $H(s)$ in partial fractions. (b) Find $H(0)$ directly from the original expression, and verify it matches the value from your decomposition. (c) A modified circuit has transfer function $\frac{5s^2 + 12s + 9}{(s + 1)(s + 2)}$. Using polynomial division, express this in the form (constant) + (partial fractions).
- (a) $5s + 7 = A(s + 2) + B(s + 1)$. At $s = -1$: $A = 2$. At $s = -2$: $B = 3$. So $H(s) = \frac{2}{s + 1} + \frac{3}{s + 2}$. (b) $H(0) = \frac{7}{1 \cdot 2} = 3.5$. Decomposition at $s = 0$: $\frac{2}{1} + \frac{3}{2} = 3.5$ ✓, confirming a match. (c) Since $(s + 1)(s + 2) = s^2 + 3s + 2$ has the same degree as the numerator, divide first: $5s^2 + 12s + 9 = 5(s^2 + 3s + 2) + (-3s - 1)$, so the expression equals $5 - \frac{3s + 1}{(s + 1)(s + 2)}$. Decomposing $-\frac{3s + 1}{(s + 1)(s + 2)}$: $3s + 1 = A(s + 2) + B(s + 1)$; at $s = -1$, $A = -2$; at $s = -2$, $B = 5$. So $\frac{5s^2 + 12s + 9}{(s + 1)(s + 2)} = 5 - \frac{2}{s + 1} + \frac{5}{s + 2}$, giving the final result $5 + \frac{2}{s + 1} - \frac{5}{s + 2}$.